

Alexandr Koryachko

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SUMMARY

Senior Machine Learning Scientist with 8+ years of experience designing, deploying, and scaling production machine learning systems in financial services and digital advertising.

Expertise spans the full ML lifecycle, including experimentation, statistical modeling, feature engineering, model monitoring, and large-scale data processing with PySpark.

Proven track record of translating business problems into production-ready ML solutions while collaborating with engineers and stakeholders.

SKILLS

PROGRAMMING

Languages:

Python • PySpark • SQL • Matlab

Packages:

Pandas • Polars • Scikit-learn • PyTorch
numpy • NLTK • spaCy • mllib • Keras

Tools:

Git • Databricks • Azure • $\LaTeX$

DATA SCIENCE

Big Data • Machine Learning

Data Visualization • Model Monitoring

Statistics • Experiment Design

LINKS

LinkedIn:// [akoryachko](#)

Github:// [akoryachko](#)

Google Scholar:// [bLSiDH4AAAAJ](#)

EDUCATION

NORTH CAROLINA STATE UNIVERSITY

PHD IN ELECTRICAL ENGINEERING

Aug 2018 | Raleigh, NC

Minor in Mathematics

GPA: 4.0 / 4.0

NATIONAL RESEARCH UNIVERSITY OF ELECTRONIC TECHNOLOGY

MS IN COMPUTER SCIENCE

Jul 2009 | Moscow, Russia

Conc. in Software Engineering

GPA: 4.0 / 4.0

BS IN COMPUTER SCIENCE

Jun 2007 | Moscow, Russia

Conc. in Information Security

GPA: 3.92 / 4.0

EXPERIENCE

XOMNIA | SENIOR DATA SCIENTIST

Feb 2025 - Present | Amsterdam, Netherlands

Consulting:

- **a.s.r.**: Productionized and scaled an LLM-based claim summarization application for company-wide use.
- **VodafoneZiggo**: Identified root causes of churn model underperformance and designed the experimentation framework for evaluating uplift models.
- **Rabobank**: Designed and prototyped a model for predicting the next transaction date in periodic payment sequences, improving prediction accuracy from 18% to 48% for natural persons.

Internal activity:

- Mentoring 4 data scientists.
- Authored a **blog post** on PySpark code productionalization.

XOMNIA | DATA SCIENTIST

Apr 2023 - Feb 2025 | Amsterdam, Netherlands

Consulting at Rabobank:

- Initiated and took a leading role in the periodicity cold start project. The project aimed to predict periodicity starting with the first transaction.
- Implemented a statistical test in PySpark on Databricks to ensure performance stability of the transaction periodicity detection model.

Internal activity:

- Authored a **blog post** on data visualization that ranked among the company's top 5 most-viewed articles for three consecutive months.
- Delivered a training session on PySpark best practices in Databricks.

VERICAST | STAFF DATA SCIENTIST I

Feb 2022 - Sep 2022 | Morrisville, NC, USA

Developed automated feature engineering functionality for big data applications within a scalable machine learning execution engine on AWS. The accompanying Python library was projected to save data scientists up to 90% of the time otherwise spent writing custom SQL queries for each new client dataset.

VERICAST | DATA SCIENTIST II

Mar 2021 - Feb 2022 | Morrisville, NC, USA

Prototyped tools for a privacy centered digital ad targeting solution that relies on webpage content as opposed to user tracking for placing clients' ads:

- Extended the optimization engine for digital advertisement auction bidding decisions with the support of cost per action campaigns.
- Built a classification model for identifying web pages with brand sensitive content. The model reduced ad serving on websites with unsafe content by 80% and improved clients' satisfaction with reporting.
- Built machine learning pipelines that analyze 300 million crawled web pages to select digital ad placements that are the most relevant to clients' desired content from 10 billion options daily.

VERICAST | DATA SCIENTIST I

Dec 2018 - Mar 2021 | Morrisville, NC, USA

Designed scalable machine learning systems processing billions of daily events for digital advertising optimization

- Leveraged historical data on the outcomes of digital ad buying auctions to detect regions of flat win probability. This allowed saving about 2% on media costs by bidding at the lower side of such regions.
- Developed machine learning models for quantifying users' interests in a variety of topics by analyzing billions of daily user URL visitations with PySpark Machine Learning toolkit.
- Designed and tuned a random forest model for inventory quality prediction for digital ad auctions using Scikit-Learn.